Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 1 · SLO 2 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 1 : SLO â€“ 2 | Stack Implementation Using Array

■ I. Push Operation â€” Insert an Element

• Step 1: IF TOP = MAX-1
• PRINT OVERFLOW
• Goto Step 4
• [END OF IF]
• Step 2: SET TOP = TOP + 1
• Step 3: SET STACK[TOP] = VALUE
• Step 4: END

■ II. Pop Operation â€” Delete an Element

• Step 1: IF TOP = -1
• PRINT UNDERFLOW
• Goto Step 4
• [END OF IF]
• Step 2: SET VAL = STACK[TOP]
• Step 3: SET TOP = TOP - 1
• Step 4: END

■ III. Peek Operation

• Step 1: IF TOP = -1
• PRINT STACK IS EMPTY
• Goto Step 3
• [END OF IF]
• Step 2: RETURN STACK[TOP]
• Step 3: END